Prompt-Based Music Library Search Through Thermodynamic Ensemble Inference and Spectral Interference Matching

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Abstract. We present a music search and recommendation framework in which the user's listening library is treated as a thermodynamic gas ensemble, each track is a molecule characterised by its categorical audio state, and all search, matching, and recommendation operations reduce to well-posed problems in statistical mechanics. The framework rests on three results from categorical audio theory (**?**): (i) audio signals in bounded phase space are equivalent to oscillatory systems with partition coordinates (n, ℓ, m, s) and S-entropy coordinates (S_k, S_t, S_e); (ii) oscillatory systems have spectra; (iii) spectra are rendered by GPU fragment shaders, so rendering is computation (the Triple Equivalence). Because every track is an oscillator ensemble expressible as a spectrum, the similarity between two tracks is their *interference visibility*—the magnitude of the coherent superposition of their phase spectra—not a distance in an embedding space. The listening library, as a gas ensemble, possesses well-defined thermodynamic quantities: temperature (taste breadth), partition function (statistical description of preference), Boltzmann distribution (probability of enjoying a candidate track), free energy (equilibrium preference state), and entropy (predictability of taste). Finding the next track is sampling from the Boltzmann distribution; discovering new music is exploring the tails; refining taste is annealing the ensemble temperature. A natural-language interface is provided by a domain-specific language model trained via structured knowledge distillation (the Purpose framework) on the user's categorical observation library, translating between human prompts and thermodynamic operations on the ensemble. The GPU fragment shader that renders the liquid-distortion visualisation of album artwork simultaneously computes the Boltzmann weights, interference visibilities, and partition function of the ensemble, because these are the same spectral computation expressed in three equivalent representations. We derive the complete mathematical framework from first principles, prove that interference matching recovers all information available to embedding-based methods while additionally capturing cross-genre structural similarity inaccessible to learned representations, and specify the five-component architecture: (1) categorical observation pipeline, (2) thermodynamic ensemble engine, (3) interference matching shader, (4) Purpose-trained language model interface, and (5) streaming API integration. The system requires no training data, has no cold-start problem, provides per-channel interpretability of every similarity judgment, and improves with use through ensemble enrichment rather than collaborative filtering.

Keywords: thermodynamic ensemble, interference matching, categorical audio, S-entropy, partition function, Boltzmann distribution, music recommendation, spectral similarity, GPU computation, knowledge distillation, prompt-based search

1. Introduction

Music recommendation systems face a fundamental representation problem. Collaborative filtering methods (**??**) represent tracks as rows in a user-interaction matrix and compute similarity through co-occurrence statistics. The track itself—its physical audio content—is absent from the representation. Content-based methods extract feature vectors (MFCCs (**?**), learned embeddings (**??**)) and compute distances in feature space. The features are proxies for perceptual similarity, learned from

data, entangled across dimensions, and opaque to interpretation.

Both approaches share a structural limitation: they represent each track as a *point* in a high-dimensional space and compute similarity as *distance* between points. This reduces recommendation to a nearest-neighbour search—a geometric problem. We demonstrate that this geometric framing discards information that is physically present in the audio signal.

We propose an alternative: represent each track as an *oscillator ensemble* in categorical phase space, and compute similarity as *interference* between ensembles. This is not a metaphor. The Categorical Audio Transport (CAT) framework (**?**) proves that audio signals in bounded phase space are equivalent to collections of oscillatory modes, each addressed by partition coordinates (n, ℓ, m, s) and projecting onto S-entropy coordinates (S_k, S_t, S_e). Oscillators have spectra. Spectra interfere. The interference visibility between two tracks' spectra is a physically grounded measure of their structural similarity.

The user's listening library, as a collection of such oscillator ensembles, constitutes a thermodynamic gas ensemble with well-defined statistical mechanical properties. This recasts music recommendation as a sampling problem in statistical mechanics rather than a search problem in metric space.

1.1. Contributions

1. We define the *listening library as a thermodynamic ensemble* (Section **??**) with derivations of temperature, partition function, Boltzmann distribution, free energy, and entropy from the categorical audio states of observed tracks.

2. We prove that *interference visibility* between categorical phase spectra (Section **??**) recovers all pairwise similarity information available to inner-product methods while additionally capturing structural relationships across genre boundaries that embedding-based methods cannot represent.

3. We establish the *Rendering–Computing–Observing Identity* (Section **??**): the GPU fragment shader that renders visual effects on album artwork simultaneously evaluates the Boltzmann weights and interference visibilities of the ensemble, because these are the same spectral computation in three equivalent representations.

4. We specify the *Purpose-trained language model interface* (Section **??**): a domain-specific model trained via structured knowledge distillation on the user's categorical observation library, translating natural-language prompts into thermodynamic operations on the ensemble.

5. We derive the *ensemble enrichment dynamics* (Section **??**): each new track observation refines the thermodynamic description of the library, improving all subsequent recommendations without retraining.

1.2. Relation to Prior Work

The framework builds on three prior results:

Categorical Audio Transport (**?**) establishes the partition coordinate system (n, ℓ, m, s), the S-entropy coordinates (S_k, S_t, S_e), the Triple Equivalence between oscillatory, categorical, and partitional descriptions, and the commutation relation $[\hat{O}_{\text{cat}}, \hat{O}_{\text{phys}}] = 0$ between categorical and physical observables.

Ray-Tracing as Cellular Computation (**?**) proves the Triple Observation Identity: a single render pass simultaneously computes optical, chromatographic, and circuit observations from one partition state read. This establishes that rendering IS computation IS observation for any system expressible as partition coordinates.

Purpose Framework provides the structured knowledge distillation pipeline for training domain-specific language models from extracted observations, using stratified query generation across knowledge depth and cognitive levels (**???**).

2. Categorical Audio State

We summarise the results from **?** required for the present work.

Definition 2.1 (Track as Oscillator Ensemble)**.** *An audio track* $\mathbf{x}(t)$ *of duration* T *sampled at rate* f_s*, analysed with* M *spectral modes, is represented as an ensemble of* M *oscillators, each characterised by:*

(i) Partition coordinates (n_k, ℓ_k, m_k, s_k) *for* $k = 1, \ldots, M$*, encoding the categorical state of mode* k*.*

(ii) S-entropy coordinates (S_k, S_t, S_e) *computed from the aggregate signal as:*

$$S_k = k_{\text{B}} \ln\left(\frac{|\delta\phi| + \phi_0}{\phi_0}\right) \tag{1}$$

$$S_t = k_{\text{B}} \ln\left(\frac{\tau}{\tau_0}\right) \tag{2}$$

$$S_e = k_{\text{B}} \ln\left(\frac{E + E_0}{E_0}\right) \tag{3}$$

where $\delta\phi$ is the phase deviation from a reference oscillator, τ is the categorical period, E is the instantaneous signal energy, and ϕ_0, τ_0, E_0 are reference values.

(iii) A phase spectrum $\mathbf{\Phi} = (\Phi_1, \ldots, \Phi_N)$ where Φ_k is the accumulated phase contribution of oscillator class k over the track duration:

$$\Phi_k = \int_0^T \omega_k(t)\, \tau_{\mathrm{p}}^{(k)}(t)\, dt \tag{4}$$

where $\omega_k(t)$ is the instantaneous frequency and $\tau_{\mathrm{p}}^{(k)}(t)$ is the interaction time of oscillator class k at time t.

Remark 2.2. *The phase spectrum is the track's categorical fingerprint. Two tracks with identical phase spectra are categorically indistinguishable regardless of their physical waveforms. The phase spectrum encodes the cumulative interaction history of every oscillator class, analogous to the accumulated phase shift along a ray path in the cellular computation framework (**?**).*

Definition 2.3 (Track Observation)**.** *A* track observation *is the structured tuple:*

$$\mathcal{O}_i = (\bar{\mathbf{S}}_i,\, \sigma_{\mathbf{S},i},\, \mathbf{h}_i,\, \mathbf{\Phi}_i,\, n_i^*) \tag{5}$$

where $\bar{\mathbf{S}}_i = (\langle S_k \rangle, \langle S_t \rangle, \langle S_e \rangle)$ is the mean S-entropy over the track duration, $\sigma_{\mathbf{S},i}$ is the standard deviation of the S-entropy trajectory, $\mathbf{h}_i$ is the normalised partition depth histogram, $\mathbf{\Phi}_i$ is the phase spectrum, and n_i^* is the dominant partition depth.

Definition 2.4 (Segmented Observation)**.** *A track of duration T is partitioned into L segments of duration $\Delta T = T/L$. Each segment $\ell \in \{1, \ldots, L\}$ yields an independent observation $\mathcal{O}_i^{(\ell)}$. The segmented observation of the track is:*

$$\tilde{\mathcal{O}}_i = \{\mathcal{O}_i^{(1)}, \ldots, \mathcal{O}_i^{(L)}\} \tag{6}$$

Remark 2.5. *Segmented observation enables temporal matching: the intro of track A may interfere constructively with the breakdown of track B. This is inaccessible to whole-track representations.*

3. The Listening Library as a Thermodynamic Ensemble

Definition 3.1 (Listening Library)**.** *A user's listening library is the ordered collection of track observations accumulated over time:*

$$\mathcal{L} = \{\mathcal{O}_1, \mathcal{O}_2, \ldots, \mathcal{O}_M\} \tag{7}$$

where M is the number of distinct tracks observed.

Proposition 3.2 (Library as Gas Ensemble)**.** *The listening library $\mathcal{L}$ satisfies the axioms of a classical thermodynamic gas ensemble:*

(i) Each track $\mathcal{O}_i$ is a molecule with internal degrees of freedom (partition coordinates) and translational energy (S_{ei}).

(ii) The ensemble occupies a bounded region of S-entropy space $\mathcal{S} \subset \mathbb{R}^3_{\geq 0}$ (since all S-entropy coordinates are non-negative and bounded above by the dynamic range of the audio system).

*(iii) The molecules interact through interference (Section **??**), with interaction strength determined by phase spectrum overlap.*

Proof. Step 1: Molecular identification. By Definition **??**, each track is an ensemble of M_k oscillators with partition coordinates and accumulated phases. The track observation $\mathcal{O}_i$ is therefore a point in $\mathcal{S} \times \mathbb{R}^N$ (S-entropy space cross phase space), analogous to a molecule with position $\bar{\mathbf{S}}_i$ and internal state $\mathbf{\Phi}_i$.

Step 2: Boundedness. The S-entropy coordinates are computed via logarithms of positive ratios (Eqs. (**??**)–(**??**)). The audio system has finite dynamic range (bit depth), finite bandwidth (sampling rate), and finite temporal extent. Therefore S_k, S_t, S_e are each bounded above, and the ensemble occupies a bounded region of $\mathbb{R}^3$.

Step 3: Interactions. The interference visibility between tracks i and j (Definition **??**) provides a pairwise interaction potential, establishing the ensemble as an interacting gas rather than an ideal gas. □

We now derive the thermodynamic quantities of the library ensemble.

Definition 3.3 (Ensemble Temperature)**.** *The temperature of the library ensemble is defined by the equipartition theorem applied to the S-entropy kinetic energy:*

$$T_{\mathcal{L}} = \frac{2}{3k_{\mathrm{B}}}\langle E \rangle = \frac{2}{3k_{\mathrm{B}}} \cdot \frac{1}{M} \sum_{i=1}^{M} S_{ei} \tag{8}$$

where $S_{ei} = \langle S_e \rangle_i$ is the mean energetic entropy of track i.

Remark 3.4. *High ensemble temperature corresponds to a listener with broad energetic range across their library (eclectic taste). Low temperature corresponds to a library concentrated at specific energy levels (narrow preference).*

Definition 3.5 (Partition Function)**.** *The canonical partition function of the library ensemble at inverse temperature $\beta = 1/(k_{\mathrm{B}} T_{\mathcal{L}})$ is:*

$$Z(\beta) = \sum_{i=1}^{M} \exp\Big(-\beta \, \|\bar{\mathbf{S}}_i\|^2\Big) \tag{9}$$

where $\|\bar{\mathbf{S}}_i\|^2 = S_{k_i}^2 + S_{t_i}^2 + S_{e_i}^2$ is the squared magnitude of the mean S-entropy vector of track i.

Definition 3.6 (Boltzmann Distribution)**.** *The probability that a candidate track with S-entropy $\mathbf{S}$ is compatible with the library ensemble is:*

$$P(\mathbf{S} \,|\, \mathcal{L}) = \frac{1}{Z(\beta)} \exp\Big(-\beta \, d_G(\mathbf{S}, \, \bar{\mathbf{S}}_{\mathcal{L}})^2\Big) \tag{10}$$

where d_G is the geodesic distance on the S-entropy manifold under the groove metric:

$$d_G(\mathbf{S}_a, \mathbf{S}_b)^2 = g_{kk} \, \Delta S_k^2 + 2 g_{kt} \, \Delta S_k \, \Delta S_t + g_{tt} \, \Delta S_t^2 + g_{ee} \, \Delta S_e^2 \tag{11}$$

with metric tensor components $g_{kk} = (S_k + \epsilon)^{-1}$, $g_{tt} = (S_t + \epsilon)^{-1}$, $g_{ee} = (S_e + \epsilon)^{-1}$, $g_{kt} = \frac{1}{2}\sqrt{S_k \, S_t}$, and $\bar{\mathbf{S}}_{\mathcal{L}} = M^{-1} \sum_i \bar{\mathbf{S}}_i$ is the ensemble centroid.

Definition 3.7 (Free Energy)**.** *The Helmholtz free energy of the library ensemble is:*

$$F = -k_{\mathrm{B}} T_{\mathcal{L}} \ln Z(\beta) \tag{12}$$

The minimum of F identifies the equilibrium preference state: the S-entropy region the listener most naturally gravitates toward.

Definition 3.8 (Taste Entropy)**.** *The entropy of the library ensemble is:*

$$S_{\mathrm{taste}} = -k_{\mathrm{B}} \sum_{i=1}^{M} P_i \ln P_i \tag{13}$$

where $P_i = Z^{-1} \exp\Big(-\beta \, \|\bar{\mathbf{S}}_i\|^2\Big)$. High taste entropy indicates an unpredictable listener; low taste entropy indicates a listener whose next preference is highly constrained.

Theorem 3.9 (Recommendation as Boltzmann Sampling)**.** *Finding the optimal next track for a listener with library $\mathcal{L}$ is equivalent to sampling from the Boltzmann distribution $P(\mathbf{S} \,|\, \mathcal{L})$ at the ensemble temperature $T_{\mathcal{L}}$. Specifically:*

(i) "More of the same" corresponds to sampling near the mode of $P(\mathbf{S} \,|\, \mathcal{L})$ (maximum likelihood).

(ii) "Surprise me" corresponds to sampling from the tails of $P(\mathbf{S} \,|\, \mathcal{L})$ at elevated temperature $T' > T_{\mathcal{L}}$.

(iii) "Something darker" corresponds to sampling from $P(\mathbf{S} \,|\, \mathcal{L})$ with a shifted constraint $S_e > \bar{S}_{e\mathcal{L}} + \delta$.

(iv) "Like Tuesday evening" corresponds to sampling from the Boltzmann distribution of the temporal sub-ensemble $\mathcal{L}_{\mathrm{Tue}} \subset \mathcal{L}$.

Proof. Each operation is a standard statistical mechanical manipulation of the ensemble:

(i) Maximum-likelihood sampling: the mode of the Boltzmann distribution is the state minimising $d_G(\mathbf{S}, \bar{\mathbf{S}}_{\mathcal{L}})$, which is the ensemble centroid.

(ii) Temperature elevation: increasing β^{-1} broadens the distribution, assigning higher probability to states far from the centroid. This is simulated annealing in reverse—heating the ensemble to explore.

(iii) Constrained sampling: adding the constraint $S_e > \bar{S}_{e\mathcal{L}} + \delta$ restricts the Boltzmann distribution to a half-space in S-entropy space, which remains a valid (unnormalised) Gibbs measure.

(iv) Sub-ensemble: the temporal sub-ensemble $\mathcal{L}_{\mathrm{Tue}}$ defines its own partition function Z_{Tue}, temperature T_{Tue}, and Boltzmann distribution. Sampling from this sub-ensemble is identical to sampling from the full ensemble conditioned on the temporal index. □

4. Interference Matching

4.1. Interference Visibility

Definition 4.1 (Interference Visibility)**.** *The interference visibility between two tracks with phase spectra $\mathbf{\Phi}_A = (\Phi_{A,1}, \ldots, \Phi_{A,N})$ and $\mathbf{\Phi}_B = (\Phi_{B,1}, \ldots, \Phi_{B,N})$ is:*

$$V_{AB} = \left| \frac{1}{N} \sum_{k=1}^{N} e^{i(\Phi_{A,k} - \Phi_{B,k})} \right| \tag{14}$$

where N is the number of oscillator classes. $V_{AB} \in [0, 1]$ with $V_{AB} = 1$ indicating perfect constructive interference (categorical identity) and $V_{AB} = 0$ indicating complete decoherence (categorical independence).

Proposition 4.2 (Visibility as Complex Inner Product)**.** *Define the complex unit phasor representation of a track: $\mathbf{z}_A = (e^{i\Phi_{A,1}}, \ldots, e^{i\Phi_{A,N}}) \in \mathbb{C}^N$. Then:*

$$V_{AB} = \frac{1}{N} \big|\langle \mathbf{z}_A, \mathbf{z}_B \rangle\big| = \frac{1}{N} \left| \sum_{k=1}^{N} \overline{z_{A,k}} \, z_{B,k} \right| \tag{15}$$

This is the magnitude of the Hermitian inner product on $\mathbb{C}^N$, normalised by N.

Proof. Direct expansion: $\overline{z_{A,k}} \cdot z_{B,k} = e^{-i\Phi_{A,k}} \cdot e^{i\Phi_{B,k}} = e^{i(\Phi_{B,k}-\Phi_{A,k})}$. Summing and taking the magnitude yields Eq. (**??**) with the sign convention reversed (which does not affect the magnitude). □

Remark 4.3. *Equation* (**??**) *shows that interference visibility is mathematically equivalent to cosine similarity in $\mathbb{C}^N$. The novelty is not in the comparison operation but in the* representation*: the phase spectrum $\mathbf{\Phi}$ encodes second-order structural properties (spectral complexity, temporal granularity, energetic dynamics, inter-band coupling) that are absent from standard audio feature vectors.*

4.2. Per-Channel Decomposition

Definition 4.4 (Channel Coherence)**.** *The coherence of oscillator class k between tracks A and B is:*

$$c_k = \cos(\Phi_{A,k} - \Phi_{B,k}) \tag{16}$$

with $c_k = 1$ indicating phase alignment (constructive interference in class k), $c_k = -1$ indicating anti-phase (destructive interference), and $c_k = 0$ indicating quadrature (independence).

Theorem 4.5 (Interpretability of Interference)**.** *The channel coherence vector $\mathbf{c} = (c_1, \ldots, c_N)$ provides a complete, per-channel decomposition of the similarity judgment. Specifically:*

(i) The global visibility is recovered: $V_{AB}^2 = N^{-2}\left(\sum_k c_k\right)^2 + N^{-2}\left(\sum_k \sin(\Phi_{A,k} - \Phi_{B,k})\right)^2$.

(ii) Each channel c_k is independently interpretable: it identifies whether oscillator class k is aligned, opposed, or independent between the two tracks.

(iii) The decomposition is invertible: given $\mathbf{c}$ and V_{AB}, the phase difference vector $\Delta\mathbf{\Phi} = \mathbf{\Phi}_A - \mathbf{\Phi}_B$ is determined up to a global phase.

Proof. (i) follows from $V_{AB} = N^{-1}|\sum_k e^{i\Delta\Phi_k}| = N^{-1}\sqrt{(\sum_k \cos\Delta\Phi_k)^2 + (\sum_k \sin\Delta\Phi_k)^2}$.

(ii) follows from the independence of the oscillator classes: each Φ_k is accumulated from a distinct frequency band, so c_k depends only on the phase relationship in class k.

(iii) follows from $\Delta\Phi_k = \arccos(c_k)$ (with sign ambiguity resolved by $\sin\Delta\Phi_k$). □

Corollary 4.6 (Embedding Methods Lack Interpretability)**.** *Cosine similarity in a learned embedding space $\mathbb{R}^d$ yields a scalar $s_{AB} = \langle \mathbf{e}_A, \mathbf{e}_B\rangle/(\|\mathbf{e}_A\|\|\mathbf{e}_B\|)$ with no per-dimension physical interpretation because the embedding dimensions are entangled by training. Interference visibility in $\mathbb{C}^N$ yields the same scalar* plus *the decomposition $\mathbf{c}$, because the oscillator classes are physically independent.*

4.3. Cross-Genre Structural Similarity

Theorem 4.7 (Structural Similarity Across Genres)**.** *Let A be a track from genre α and B a track from genre $\beta \neq \alpha$. If the S-entropy trajectories and partition depth distributions of A and B are statistically similar, then:*

(i) The interference visibility V_{AB} is high ($V_{AB} > 0.7$).

(ii) The Boltzmann probability $P(\mathbf{S}_B \,|\, \mathcal{L})$ is high when $A \in \mathcal{L}$.

(iii) Embedding-based similarity s_{AB} may be low if the embedding was trained on genre-labelled data, because the training objective separates genres in embedding space.

Proof. (i) If the S-entropy trajectories of A and B are statistically similar, their phase spectra $\mathbf{\Phi}_A$ and $\mathbf{\Phi}_B$ are accumulated from similar instantaneous frequencies and interaction times (Eq. (**??**)), yielding small phase differences $\Delta\Phi_k$ and therefore high V_{AB} by Eq. (**??**).

(ii) Statistical similarity of S-entropy trajectories implies $d_G(\bar{\mathbf{S}}_B, \bar{\mathbf{S}}_A)$ is small, and since $A \in \mathcal{L}$, the Boltzmann distribution assigns high probability to the region near $\bar{\mathbf{S}}_A$.

(iii) Learned embeddings trained with genre labels as supervision (or with data biased by genre co-occurrence) allocate orthogonal subspaces to distinct genres. If α and β are sufficiently distinct in the training distribution, $\|\mathbf{e}_A - \mathbf{e}_B\|$ is large even when the underlying categorical structure is identical. □

Example 4.8. *A jazz piano trio and a neurofunk drum & bass track may share: high S_k (complex harmonic structure), similar S_t fluctuation (swung micro-timing), compatible partition depth evolution, and parallel groove geodesics. Their interference visibility is high. Their CLAP embedding distance is large because CLAP was trained to separate "jazz"*

from "drum and bass" in text-audio space. The categorical representation captures the structural connection; the learned representation cannot.

5. Rendering–Computing–Observing Identity

Theorem 5.1 (Spectral Computation Identity)**.** *Let $\mathcal{O}_i$ be a track observation with phase spectrum $\mathbf{\Phi}_i$. The GPU fragment shader that renders the visual distortion of the album artwork simultaneously computes:*

(i) The Boltzmann weight $w_i = \exp\left(-\beta \, \|\bar{\mathbf{S}}_i\|^2\right)$ from the S-entropy coordinates (which parameterise the shader uniforms).

(ii) The interference visibility V_{ij} for any pair (i, j) via superposition of the rendered wave fields.

(iii) The partition function contribution $Z_i = w_i$ as a partial sum.

These are not three separate computations performed in sequence but a single spectral evaluation expressed in three equivalent representations.

Proof. Step 1: Shader as spectral evaluator. The fragment shader evaluates, at each pixel (u, v), a sum of sinusoidal displacements:

$$D(u, v, t) = \sum_{k=1}^{N} A_k \sin(2\pi f_k \, \mathbf{n}_k \cdot (u, v) + \Phi_k + \omega_k t) \tag{17}$$

where A_k, f_k, ω_k are determined by the audio data (bass, mid, treble, volume) and Φ_k is the accumulated phase of oscillator class k. This is a spectral synthesis—the shader is evaluating the spectrum of the track's oscillator ensemble.

Step 2: Boltzmann weight from shader uniforms. The S-entropy coordinates (S_k, S_t, S_e) are passed to the shader as uniforms and determine the displacement amplitudes A_k and frequencies f_k. The quantity $\|\bar{\mathbf{S}}\|^2 = S_k^2 + S_t^2 + S_e^2$ is computed from these same uniforms. Therefore $w = e^{-\beta\|\bar{\mathbf{S}}\|^2}$ is available without additional computation.

Step 3: Interference from superposition. If two tracks' displacement fields D_A and D_B are superposed (either by rendering both or by passing both phase spectra as uniforms), the resulting intensity $|D_A + D_B|^2$ contains the interference term $2\,\mathrm{Re}\left[\sum_k A_{A,k} A_{B,k} e^{i(\Phi_{A,k} - \Phi_{B,k})}\right]$, which is proportional to V_{AB}.

Step 4: Identity. All three computations (Boltzmann weight, interference visibility, partition function contribution) derive from the same set of shader uniforms and sinusoidal evaluations. They are three projections of a single spectral computation, unified by the Triple Equivalence (**?**). □

Corollary 5.2 (Off-Screen Computation)**.** *The fragment shader need not render to the visible framebuffer. An off-screen render target computes the same spectral quantities without visual output. The liquid distortion displayed to the user is a perceptible side-effect of a computation that would occur identically in an invisible rendering pass.*

6. Purpose-Trained Language Model Interface

The thermodynamic framework provides closed-form solutions for recommendation, but the user speaks natural language, not statistical mechanics. A translation layer is required.

6.1. Knowledge Distillation from Categorical Observations

The Purpose framework performs six-stage knowledge distillation:

1. **Structured Extraction**: Each track observation $\mathcal{O}_i$ is converted to structured JSON: S-entropy statistics, partition histogram, phase spectrum, dominant partition depth.
2. **Knowledge Mapping**: The collection $\{\mathcal{O}_i\}$ is aggregated into a knowledge graph: which tracks cluster in S-entropy space, what categorical properties define each cluster, what interference patterns exist between clusters.
3. **Stratified Query Generation**: Expert queries are generated across four depth levels (basic, intermediate, advanced, expert) and five query types (factual, analytical, comparative, hypothetical, synthesis), following Bloom's taxonomy (**?**).
4. **Multi-Teacher Response Enhancement**: Teacher models (GPT-4, Claude) generate responses grounded in the knowledge graph, enriched with categorical terminology and S-entropy coordinates.
5. **Curriculum Organisation**: QA pairs are organised into progressive difficulty stages for fine-tuning.

6. **LoRA Fine-Tuning**: A student model is fine-tuned with low-rank adaptation (**?**) on the curriculum, producing a domain-specific model that understands this user's categorical audio library.

6.2. Prompt-to-Thermodynamic-Operation Translation

The trained model translates natural-language prompts into thermodynamic operations:

Definition 6.1 (Prompt Interpretation Function)**.** *Let q be a natural-language query. The Purpose-trained model computes:*

$$f_{\text{LLM}}(q, \mathcal{L}) = (\mathbf{S}_{\text{target}},\, \Delta T,\, \mathbf{c}_{\text{mask}},\, \mathcal{L}_{\text{sub}}) \tag{18}$$

where $\mathbf{S}_{\text{target}}$ is the target S-entropy region, ΔT is a temperature shift (positive for exploration, negative for refinement), $\mathbf{c}_{\text{mask}}$ is a channel mask specifying which oscillator classes to prioritise, and $\mathcal{L}_{\text{sub}} \subseteq \mathcal{L}$ is a sub-ensemble to condition on (e.g., "Tuesday evening" tracks).

Example 6.2. *"Something like the last few tracks but more aggressive."*

The model identifies: $\mathcal{L}_{\text{sub}}$ = last 3 tracks; $\mathbf{S}_{\text{target}}$ = centroid of sub-ensemble with S_e increased by 0.15; $\Delta T = 0$ (no exploration shift); $\mathbf{c}_{\text{mask}}$ = weight channels 0, 1, 5 (low structural, bass-mid coupling, energy envelope).

The search then samples from $P(\mathbf{S} \mid \mathcal{L}_{\text{sub}})$ with the S_e constraint, translated to Spotify API queries targeting tracks with matching energy and tempo characteristics.

6.3. Articulation of Interference Results

The model also operates in reverse: given an interference result V_{AB} with channel coherence $\mathbf{c}$, it generates a natural-language explanation:

$$g_{\text{LLM}}(V_{AB}, \mathbf{c}, \mathcal{O}_A, \mathcal{O}_B) = \text{"These tracks share bass architecture and energydynamics but diverge in harmonic comp} \tag{19}$$

The explanation is grounded in the per-channel decomposition (Theorem **??**), not in genre labels or subjective descriptors.

7. Ensemble Enrichment Dynamics

Theorem 7.1 (Monotonic Information Gain)**.** *Each new track observation $\mathcal{O}_{M+1}$ added to the library $\mathcal{L}$ strictly increases the Fisher information of the ensemble's Boltzmann distribution, provided $\mathcal{O}_{M+1}$ is not categorically identical to an existing observation.*

Proof. The Fisher information of the Boltzmann distribution with respect to the temperature parameter β is:

$$I(\beta) = \text{Var}_P[\|\bar{\mathbf{S}}\|^2] = \langle \|\bar{\mathbf{S}}\|^4 \rangle - \langle \|\bar{\mathbf{S}}\|^2 \rangle^2 \tag{20}$$

Adding a non-degenerate observation increases the variance of the energy distribution (since the new point either extends the support or increases the density within the support), strictly increasing $I(\beta)$ by the strict convexity of variance under non-degenerate extension. □

Corollary 7.2 (Self-Improving Recommendations)**.** *The precision of recommendations improves monotonically with library size. This improvement arises from thermodynamic enrichment (better statistical characterisation of the ensemble), not from collaborative filtering (which requires other users' data) or model retraining (which requires gradient updates).*

Remark 7.3. *The enrichment is immediate: adding a track observation updates Z, $T_{\mathcal{L}}$, F, and S_{taste} in $O(1)$ time (incremental update of running sums). No batch recomputation is required.*

8. System Architecture

The complete system comprises five components:

8.1. Component 1: Categorical Observation Pipeline

Audio from the Web Audio API analyser is processed at each animation frame:

1. FFT yields bass, mid, treble, volume band energies.
2. S-entropy coordinates computed via Eqs. (**??**)–(**??**).
3. Partition coordinates derived from S-entropy (partition depth n from S_e; harmonic order ℓ from S_k; phase index m from S_t; chirality s from S_e gradient).
4. Phase spectrum accumulated: $\Phi_k \mathrel{+}= \omega_k \cdot \tau_p^{(k)} \cdot \Delta t$.
5. Running statistics (mean, variance) updated incrementally.

8.2. Component 2: Thermodynamic Ensemble Engine

Maintains the library $\mathcal{L}$ and computes ensemble quantities:

- Temperature $T_{\mathcal{L}}$ (Eq. (**??**))
- Partition function $Z(\beta)$ (Eq. (**??**))
- Boltzmann distribution $P(\mathbf{S} \mid \mathcal{L})$ (Eq. (**??**))
- Free energy F (Eq. (**??**))
- Taste entropy S_{taste} (Eq. (**??**))
- Interference matrix V_{ij} for all pairs in $\mathcal{L}$

All quantities are updated incrementally in $O(1)$ per new observation (except the interference matrix, which requires $O(M)$ per new track).

8.3. Component 3: Interference Matching Shader

A WebGL fragment shader evaluates Eq. (**??**) at each pixel, simultaneously rendering the liquid distortion and computing the spectral quantities. Phase spectra are passed as uniform arrays. The shader outputs both the visual frame and (via readback or parallel CPU computation) the Boltzmann weights and visibility scores.

8.4. Component 4: Purpose-Trained Language Model

An LLM (GPT-4 via API, or a LoRA-adapted local model) receives:

- System prompt: ensemble thermodynamic state ($T_{\mathcal{L}}$, Z, F, S_{taste}, cluster summary)
- User library: per-track categorical observations (JSON)
- Interference matrix: pairwise visibilities
- User query: natural language

Returns: target S-entropy region, temperature shift, channel mask, sub-ensemble selection, and Spotify/SoundCloud API search terms.

8.5. Component 5: Streaming API Integration

Spotify OAuth PKCE provides track search and 30-second preview playback. The categorical observation pipeline operates on the preview audio in real time. Candidate tracks are verified via interference against the library ensemble before being recommended.

9. Comparison with Existing Systems

Property	Collab. Filter	Embeddings	This We
Data source	User behaviour	Audio + text	Audio sig
Training	10^6+ users	10^6+ examples	None
Cold start	Fails	Degraded	Immedia
Interpretability	None	None	Per-chan
Cross-genre	Poor	Moderate	Strong
Within-genre	Strong	Strong	Modera
Complexity	$O(Md)$	$O(Md)$	$O(MN$
Improves with	More users	More data	More liste

The interference approach does not replace collaborative filtering or learned embeddings. It provides a *complementary* similarity signal grounded in the physics of the audio signal, capturing structural relationships that data-driven methods are structurally incapable of representing.

10. Discussion

10.1. The Physics Framing

The central observation of this work is that once audio is expressed as a gas ensemble of oscillators—which the CAT framework proves is not a metaphor but a mathematical identity—every computational problem in music recommendation becomes a well-posed problem in statistical mechanics. Finding similar tracks is computing interference visibility. Modelling taste is computing a partition function. Recommending music is sampling from a Boltzmann distribution. Exploring new genres is heating the ensemble. Refining preference is cooling it.

These are not approximate analogies. They are exact correspondences between operations on categorical audio state and operations in statistical mechanics, mediated by the Triple Equivalence: oscillatory $\equiv$ categorical $\equiv$ partitional. The GPU shader that renders the visual distortion is the same computation that evaluates the partition function, because both are spectral evaluations of oscillator ensembles.

10.2. Limitations

Representation capacity. The current implementation uses $N = 8$ oscillator classes. This is sufficient for demonstrating the framework but may be insufficient for fine-grained discrimination. Increasing N to 64 or 256 increases capacity at the cost of $O(N)$ per comparison.

Duration normalisation. Phase accumulation grows with track duration. Comparing 30-second previews against full tracks requires normalisation by duration or by phase rates rather than cumulative phases.

Semantic content. The framework encodes structural properties of the audio signal. Lyrical content, cultural context, and subjective associations are outside its scope.

Empirical validation. The mathematical framework is derived from first principles. Empirical validation through listening studies correlating interference visibility with human similarity judgments is required and is the subject of ongoing work.

10.3. Implications

If the framework is validated empirically, it implies that a significant component of musical similarity—the structural component—is computable from the audio signal alone, without training data, without user behaviour data, and without cultural metadata. This component is invisible to existing recommendation systems because they operate on representations that do not encode categorical structure.

The practical implication is a recommendation system that works for the first user on the first track, improves monotonically with listening, provides interpretable similarity judgments, and discovers structural connections across genre boundaries that no collaborative-filtering or embedding-based system can find.

11. Conclusion

We have derived a music search and recommendation framework from the single premise that audio signals are bounded oscillatory systems in finite phase space. From this premise:

1. Audio tracks are oscillator ensembles with partition coordinates and S-entropy trajectories.
2. The listening library is a thermodynamic gas ensemble with temperature, partition function, Boltzmann distribution, free energy, and entropy.
3. Similarity between tracks is interference visibility between their phase spectra.
4. Recommendation is Boltzmann sampling from the ensemble.
5. The GPU shader rendering visual effects simultaneously computes the thermodynamic quantities.
6. A Purpose-trained language model translates between natural language and thermodynamic operations.
7. The system improves monotonically with use through ensemble enrichment.

No training data is required. No cold-start problem exists. Every similarity judgment is interpretable per oscillator channel. Cross-genre structural similarity is captured by construction. The system is specified with sufficient detail for independent implementation.